GEOINT ASSESSMENT

Rocket Lab Launch Complex-1 (LC-1), Mahia Peninsula, New Zealand

Operational Capacity Evaluation

Mahia Peninsula, New Zealand

Assessment type	GEOINT assessment — operational capacity evaluation
Launch complex	Rocket Lab Launch Complex-1 (LC-1), Mahia Peninsula, New Zealand
Launch vehicle	Electron
Launch pads	2 (LC-1A, LC-1B)
Estimated maximum sustainable capacity	40–55 launches per year (moderate confidence)
Observed turnaround	~14 days per pad; technical minimum ~48 hours
Highest observed annual total	21 launches in 2025
Rocket production	Rocket Lab Auckland manufacturing facility, New Zealand (off-site)
Source basis	Commercial satellite imagery (Google Earth and equivalent platforms); public launch records and mission timelines; Rocket Lab official statements and investor materials; industry media and independent spaceflight tracking

RESEARCH QUESTION

What is the maximum sustainable annual launch throughput of LC-1, and which factors constrain it?

Prepared from public sources only. No private, paid or non-public data was used.

Executive Summary (Key Judgments)

- Estimated sustainable maximum capacity: 40-55 launches per year (moderate confidence)
- Two independent launch pads (LC-1A, LC-1B) enable parallel operations
- Observed operational turnaround suggests ~26 launches per pad annually
- Primary system constraint: rocket manufacturing and delivery, not launch infrastructure
- Current activity levels remain below infrastructure capacity
- Operational activity is highly observable ; significant undeclared capacity is unlikely

IMPLICATION
Schedule risk is primarily driven by production variability rather than launch site limitations.

Client Context

This assessment was conducted to support a commercial evaluation of launch availability, schedule reliability, and operational scalability for small satellite missions. The analysis estimates the maximum sustainable operational capacity of Launch Complex-1 based on:

- satellite imagery (GEOINT)
- open-source reporting (OSINT)
- observed operational patterns
- system-level infrastructure logic

Historical launch data is used as a baseline for realism , not as a direct measure of capacity.

Analytical Objective

Key question: What is the maximum sustainable annual launch throughput of LC-1, and which factors constrain it? The assessment evaluates:

- launch infrastructure structure and scale
- ability to conduct parallel operations
- minimum sustainable turnaround time
- system bottlenecks
- production and logistics dependencies
- operational transparency
- likelihood of undisclosed capacity

1. Launch Infrastructure

Satellite analysis and open-source confirmation indicate two active launch pads : LC-1A and LC-1B. Observations:

- Spatial separation between pads
- Independent access routes
- Separate safety exclusion areas
- No observable shared critical infrastructure limiting independent use

This configuration indicates the capability for parallel launch preparation and execution . Infrastructure scale and layout are compact and centralized, with no observable excess or reserve launch capacity .

Assessment
Launch infrastructure supports parallel operations with no identified pad-related bottlenecks.

Confidence: High

2. Minimum Launch Interval

Confirmed operational data indicates that in 2025 the site supported two launches within less than 48 hours . This is assessed as short-term surge capability , not a sustainable operating tempo. Observed operational patterns show:

- Typical intervals of approximately two weeks
- Periods of up to four launches per month

Estimated sustainable turnaround per pad: 365 / 14 ≈ 26 launches per year

Assessment

Technical minimum ~48 hours; sustainable operational tempo ~26 launches annually per pad.

Confidence: Moderate

3. Integration and Processing

Electron rocket production and final assembly occur off-site at Rocket Lab's Auckland manufacturing facility. Imagery of LC-1 shows:

- Limited processing infrastructure
- No large multi-bay integration complexes

This configuration indicates the launch site functions primarily as an operational processing and launch location , with manufacturing externalized.

> SYSTEM IMPLICATION
> Overall launch rate depends on production throughput and delivery cadence.

Assessment

Primary system bottleneck is likely external manufacturing capacity rather than launch infrastructure.

Confidence: High

4. Logistics and Site Context

LC-1 is located on a remote coastal peninsula with:

- Single primary road access
- No industrial clusters within ~5 km
- Surrounding land primarily agricultural

Infrastructure is compact and centralized. No distributed storage or auxiliary operational zones were observed. For a small-class launch vehicle, the logistics model appears sufficient and not capacity-limiting . No observable indicators of:

- hidden operational areas
- additional launch positions
- distributed support infrastructure

Confidence: Moderate-High

5. Production Capacity

Rocket Lab has publicly stated long-term production goals of approximately one Electron per week (~52 per year) . Actual performance:

- 21 launches in 2025 (highest observed annual total)

This indicates current production supports >20 launches annually but remains below stated targets. Comparison with launch infrastructure capacity suggests production rate will determine achievable system throughput .

Confidence: Moderate

6. Historical Launch Activity

Table 1. Electron launches by year, as stated in the source

Year	Launches
2022	9
2023	10
2025	21

Figure 1. Electron launches by year

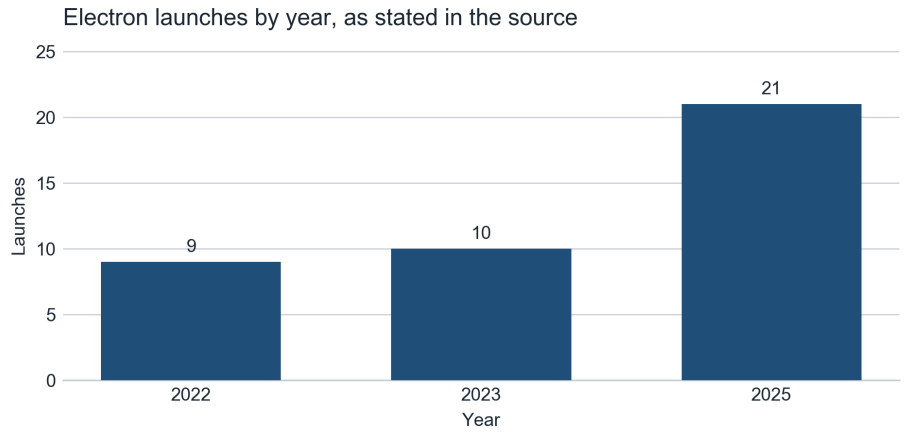


Chart of the annual launch counts stated in the source: 2022 - 9, 2023 - 10, 2025 - 21. The source lists no figure for 2024 and no bar is shown for that year; the gap is not a zero. These are observed launch counts, not a capacity estimate.

2025 21 Trend indicates increasing operational utilization , though current activity remains well below calculated infrastructure capacity.

Confidence: High

7. Operational Transparency and Observability

Operational activity is highly visible:

- Public launch announcements
- Live mission broadcasts
- Independent tracking by industry media and analytical communities

Given:

- small infrastructure footprint
- limited number of launch positions
- high public visibility

The existence of significant undisclosed operational capacity is assessed as unlikely .

Assessment

Operational activity is highly transparent; substantial discrepancies between actual and declared activity would likely be detected.

Confidence: High

8. System Capacity Model

Parameters:

- Launch pads: 2
- Sustainable turnaround: ~14 days

Calculation

≈ 26 launches per pad annually ≈ 52 launches per year theoretical sustainable maximum

Real-world adjustments

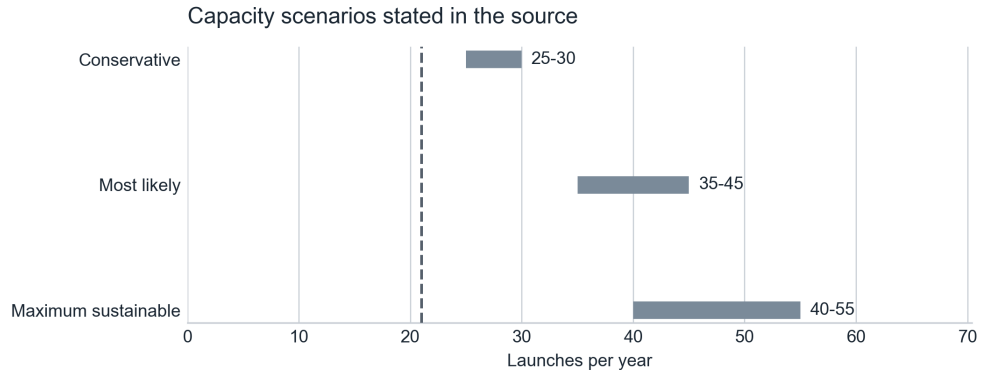
- production constraints
- weather and operational disruptions
- demand variability

Capacity Scenarios

Table 2. Capacity scenarios, as stated in the source

Scenario	Annual Launches
Conservative	25-30
Most likely	35-45
Maximum sustainable	40-55

Figure 2. Capacity scenarios and highest observed annual total



Range bars are the annual launch-count scenarios stated in the source: Conservative 25-30, Most likely 35-45, Maximum sustainable 40-55. The single marker is the source's highest observed annual total, 21 launches in 2025. The source states no figure for 2024. Bar extent is the stated range only; no midpoint or projection has been derived.

Risk Assessment

Table 3. Risk assessment, as stated in the source

Factor	Assessment	Confidence
Launch infrastructure bottleneck	Low	High
Production constraint	Moderate	Moderate
Weather/operational delays	Moderate	Moderate
Hidden capacity risk	Low	High

Final Assessment

Estimated maximum sustainable capacity: 40-55 launches per year

Primary limiting factor: Rocket production and delivery, not launch infrastructure.

Confidence: Moderate

> TRANSPARENCY ASSESSMENT
> Operational activity is highly observable; significant deviations from declared or observed utilization are unlikely to remain undetected.

System-Level Judgment

Launch Complex-1 represents a compact, centralized commercial spaceport optimized for high-frequency small-launch operations. The presence of two independent launch pads provides excess launch capacity relative to current utilization . Available evidence indicates that system throughput is constrained primarily by manufacturing scale and operational demand , rather than by launch site infrastructure. No observable indicators suggest hidden infrastructure or additional undisclosed launch capability.

Sources (Data Types)

- Commercial satellite imagery (Google Earth and equivalent platforms)
- Public launch records and mission timelines
- Rocket Lab official statements and investor materials
- Industry media and independent spaceflight tracking resources

Analytical Limitations

- No access to internal production or scheduling data
- Weather impacts not quantitatively modeled
- Customer backlog and demand dynamics not assessed